Technical Terminology Master Primer

Zero-to-Advanced Concepts & Guide Preparation Checklist

1. The Fundamentals: AI, Machine Learning, and NLP Basics

Artificial Intelligence (AI)

The broad engineering field of making machines perform cognitive tasks (reading, reasoning, seeing). In our project: AI reads messy customer reviews, analyzes product photos, and checks consistency against seller descriptions.

Natural Language Processing (NLP)

The branch of AI dedicated to understanding human text. NLP handles sarcasm, typos, and indirect phrasing (e.g., distinguishing "looks nice, but died in 2 days" from genuine praise).

Tokenization & Tokens

Computers cannot read characters directly; they only process numbers. Tokenization chops sentences into atomic units (words or sub-words like "un-break-able") and assigns each a unique numerical ID.

2. Language Models & Transformer Foundations

Large Language Model (LLM)

A massive deep neural network (e.g., Gemini, GPT-4) trained on billions of web words. It excels at parsing unstructured text into structured formats. The PRAISE (ACL 2025) base paper uses Gemini Flash to extract attributes from reviews.

Self-Attention Mechanism (Transformer)

Instead of reading text word-by-word from left to right, Self-Attention examines all words simultaneously and calculates mathematical connection weights between every word pair. Example: In "The shoe slipped because it lacked grip", it connects "it" directly to "shoe".

Embeddings (Dense Vectors)

GPS coordinates for meaning: A list of 384 floating-point numbers representing the semantic concept of a sentence. Sentences with identical meanings ("Fits true to size" vs "Sizing is accurate") map to almost identical coordinate points in vector space.

Cosine Similarity

A geometric formula measuring the angle between two vectors: +1.0 = identical meaning; 0.0 = completely unrelated; -1.0 = direct opposite meaning.

3. Text Reasoning & Inference (Our Text Core)

Natural Language Inference (NLI)

A core NLP reasoning task testing logical entailment between a Premise (Customer Review) and a Hypothesis (Seller Claim):

- Entailment (Support): Review proves the claim is true ('cobbler confirmed it is real leather' -> '100% Leather')
- Contradiction: Review proves the claim is false ('cheap vinyl plastic, not leather' -> '100% Leather')
- Neutral: Review is related but does not prove or refute ('looks good with blue jeans' -> '100% Leather')

Bi-Encoder (Sentence-BERT) vs. Cross-Encoder (RoBERTa)

- Bi-Encoder (all-MiniLM-L6-v2): Encodes sentences independently into vectors. Comparing them takes microseconds. Used in Step 2 to scan thousands of reviews and retrieve Top-K relevant sentences.
- Cross-Encoder (nli-roberta-base): Feeds claim and review together through cross-attention layers. Slower, but dramatically more accurate for deep logical reasoning. Used in Step 3 to classify Support vs. Contradict.

Atomic Propositions / Claim Deconstruction (Tp)

Splitting a long seller paragraph into single, indivisible, testable statements with category tags (e.g., [MATERIAL], [SIZING], [PERFORMANCE]). Prevents a true claim from being obscured by a false claim.

Softmax Function

A mathematical formula at the model output converting unconstrained numbers (logits) into clean probabilities that sum to exactly 1.0 (P(Support) + P(Contradict) + P(Neutral) = 1.0).

4. Multimodal AI & Computer Vision (Our Vision Engine)

Why RGB Space Fails Human Eyes

Standard screen RGB (Red, Green, Blue) is perceptually non-linear. The human eye has different sensitivities to green, yellow, and blue. Two colors with a large numerical RGB distance might look identical, while a small blue shift can look massive.

CIE L*a*b* Color Space & CIE76 Delta E

Designed by international color scientists to be perceptually uniform: L* = Lightness (0 black to 100 white); a* = Green-to-Red; b* = Blue-to-Yellow. Delta E measures Euclidean distance in LAB space: Delta E < 10 (ambient light variance); Delta E > 25 (severe color discrepancy).

K-Means Clustering

An unsupervised machine learning algorithm grouping pixel colors into K clusters, used to extract the 3 dominant product colors while ignoring white studio backgrounds.

OpenAI CLIP (Contrastive Language-Image Pretraining)

A multimodal neural network trained on 400M image-text pairs. It maps images into semantic vectors. Passing a catalog photo and an unboxing photo through CLIP lets us calculate visual cosine similarity, verifying shape, pattern, and style independent of room lighting.

5. System Scoring & 4-Tier Taxonomy

Product Representation Index (PRI: 0-100%)

PRI = 100 * [alpha * S_mean + (1 - alpha)*(1 - C_mean) - beta * D_visual]. Synthesizes text support/contradiction and visual discrepancy into a single reliability dial. (alpha=0.6, beta=0.15).

The 4-Tier Assessment Taxonomy

1. Consistent: Strong empirical evidence validates seller representations (S >= 0.50, C < 0.30).
2. Contradictory: Customer evidence directly refutes one or more core claims (C >= 0.50 or D_visual >= 0.50).
3. Mixed Evidence: Customer feedback is bimodal or divided (common in clothing/shoe sizing debates).
4. Insufficient Evidence: Product has fewer than 3 reviews (cold-start); triggers intrinsic buzzword audit.

6. Software Tools & Infrastructure

- Hugging Face (transformers): The industry-standard open-source library hosting our pretrained AI models.
- PyTorch (torch): The deep learning framework developed by Meta AI executing tensor math under the hood.
- OpenCV (cv2): Computer vision library used for image preprocessing and RGB-to-LAB color space conversion.
- Streamlit: Python framework used to render our interactive web dashboard without complex frontend code.
- Amazon Reviews Benchmark (2024 / 2018): Real-world dataset from UCSD / Julian McAuley Lab.

7. The "Guide Meeting" Rapid-Fire Checklist (10 Core Concepts)

1. Project Core: We test if seller claims (specs, size, color) match customer reviews and unboxing photos.
2. Base Paper: PRAISE from ACL 2025 (IIIT Hyderabad, UPenn, ASU) - established review attribute comparison.
3. Our Novelty: We add Multimodal Vision (CIE LAB Delta E + CLIP), S-BERT semantic filtering, and PRI scoring.
4. What is NLI?: Natural Language Inference - determines if a review sentence logically Supports or Contradicts a claim.
5. Why S-BERT?: Fast Bi-Encoder scanning thousands of reviews in milliseconds to isolate Top-K relevant sentences.
6. Why RoBERTa?: Deep Cross-Encoder performing full cross-attention between claim and review for high-accuracy reasoning.
7. Why CIE LAB?: Perceptually uniform color space where Delta E accurately mirrors human color perception unlike RGB.
8. Why CLIP?: Measures visual geometry, silhouette, and pattern similarity between photos regardless of lighting.
9. Missing Photos?: Vision engine gracefully bypasses (beta = 0); the text consistency engine runs at 100% capacity.
10. Beneficiaries: Shoppers (pre-purchase truth), Platforms (slashing 30% fashion return costs), and Sellers (fixing errors).

CONFIDENCE SUMMARY FOR MEETING YOUR GUIDE:
You now understand every piece of the pipeline: why we deconstruct claims, how S-BERT filters noise, how RoBERTa classifies contradiction, how LAB Delta E measures color shifts, and how PRI aggregates everything. Walk into your meeting knowing you have an ACL 2025 top-tier base paper and a complete, well-engineered architecture.